

Earth, Wind and Fire: Health Outcomes of Long-Range PM_{2.5} Transport from Amazon Fires*

Julio de Barros[†] Guilherme Jardim[‡] Akiva Rube[§]

Abstract

Human-caused fires in the Amazon are a major source of fine-particulate pollution. We ask whether it reaches Brazil's South and Southeast, more than 1,000 km downwind, and harms health there. An atmospheric-transport model traces fire emissions to their destinations and condenses them into a single fire-attributable PM_{2.5} index. Ground monitors confirm the index captures real airborne particulate: a one-standard-deviation rise in predicted exposure lifts measured PM_{2.5} by 0.055 and PM₁₀ by 0.030 standard deviations, while pollutants unrelated to fire register nothing. That exposure produces an acute respiratory response. Admissions rise about half a percent per standard deviation, concentrated on the dry days when rain does not scavenge the plume, with clean placebos.

JEL codes: Q53, I18, Q54, Q23, O13.

Keywords: air pollution; fine particulate matter; Amazon fires; deforestation; atmospheric transport; hospital admissions; Brazil.

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[†]Columbia SIPA. Email: julio.m@columbia.edu

[‡]Department of Economics, PUC-Rio. Email: gnjardim1@gmail.com

[§]Yeshiva University. Email: akivarube@gmail.com

1 Introduction

The large share of global greenhouse gas (GHG) emissions coming from fire activity in the Amazon during the early 2000's, resulted in mounting international pressure on Brazil to regulate deforestation. In response to this, the federal government launched the Real-Time System for Detection of Deforestation (DETER) under the Action Plan for the Prevention and Control of Deforestation in the Legal Amazon (PPCDAm). The DETER system utilizes satellite imaging to transmit real-time alerts of deforestation to environmental law enforcement. It is estimated that the combination of real-time alerts with the rapid deployment of law enforcement, lowered deforestation by 59% in the years following implementation of the policy compared to the scenario without the policy (Assunção et al., 2014).

This pattern, however, seems to have changed. The 2020 fire season in the Brazilian Amazon exceeded that of 2019 and both years were a clear departure from the post-2012 trend of much lower intensity fires recorded by the VIIRS sensor on board the Suomi NPP satellite mission (Global Fire Emissions Database, 2020a). The full fires time series, going back to 1997, shows that the recent positive trend in fires is similar to the early 2000s period when the most intense fires and aggressive deforestation activities were registered (Global Fire Emissions Database, 2019). A new model that characterizes fires by satellite-observable characteristics such as perimeter, fire radiative power, and persistence, confirms that increased deforestation is the main cause of the recent fire activity in the Amazon (Global Fire Emissions Database, 2020b; Andela et al., 2019). The 2019 fire season, for instance, was characterized by large and intense occurrences near roads, an indication of land clearing fires (Barber et al., 2014; Voiland, 2019). The media and NGOs have been reporting¹ that the PPCDAm's enforcement mechanisms are no longer operational, which has led to the observed increase in deforestation.

In August 2019, dramatic images of smoke from fires in the Amazon arriving in cities nearly 2000 miles away, brought the scale of deforestation into the public eye (Humans Rights Watch, 2020; Faiola et al., 2019). While much of the attention surrounding these fires has been about the need to preserve the Amazon as a tropical rainforest and to prevent it from transitioning to a savannah-like biome, there is also concern for the

¹See, for instance, the coverage by [WWF](#) and [The New York Times](#).

immediate and substantial release of particulate matter into the atmosphere, as this has the potential to significantly alter the quality of air in nearby cities. To this end, Rocha and Sant’Anna (2020) have looked at the local health consequences of fire generated pollution in the Amazon, and uncovered a significant causal link with hospitalizations due to respiratory conditions. We extend on their work by exploring whether the health impacts of these fires can be detected as far as the South and Southeast of Brazil, a region that is 1,800 miles south of the Amazon Rainforest and that contains the largest portion of the Brazilian population.

In order to reliably assess the long range health impacts of fires in the Amazon Rainforest, we use data on the location of fires in this region from Moderate Resolution Imaging Spectroradiometer Collection 6 (MODIS C6) in combination with the Hybrid Single-Particle Lagrangian Integrated Trajectory model (HYSPLIT), an atmospheric transport and dispersion model that offers hourly resolution air parcel trajectories. Our reliance on the model allows us to assign pollution exposure to municipalities in a manner that should not correlate with other determinants of health. In addition to this, we are able to isolate other sources of pollution from our estimates, identifying the exact effect of pollutant emission from fires in the Brazilian Amazon. Ground monitor readings of particulate matter confirm the pollution outputs we generate with the atmospheric transport model.

The model assigns to each fire the black carbon (BC) and organic carbon (OC) concentration at its source pixel and transports these species downwind. Because the health literature speaks in terms of fine particulate matter rather than its carbonaceous constituents, we combine BC and OC into a single fire-attributable $PM_{2.5}$ index, which we label *FirePM25*, using a standard atmospheric-chemistry bridge that maps primary carbonaceous emissions into total fine particulate mass (Section 3). This index is the natural exposure measure for a health study, and it is the quantity whose magnitude a policy audience can interpret directly. We report it alongside its BC and OC components throughout, so that the compositional information is preserved.

Our results indicate that exposure to $PM_{2.5}$ generated by fires in the Amazon rainforest is detected far downwind and is associated with respiratory hospital admissions in the municipalities of the South and Southeast of Brazil. First, the HYSPLIT-predicted

FirePM25 index that reaches São Paulo loads positively and very precisely on the $\text{PM}_{2.5}$ and PM_{10} recorded by ground monitors, and shows no relationship with pollutants unrelated to biomass burning (NO and SO_2), confirming that the model captures fire-driven particulate matter and not other sources. Second, estimating Poisson regressions on daily municipality admission counts, fire-driven $\text{PM}_{2.5}$ raises respiratory admissions on the dry days when the arriving plume is not scavenged by rain: the implied dry-day effect of a one-standard-deviation increase in the FirePM25 index is about half a percent, and it is concentrated on respiratory rather than cardiovascular conditions. Third, this respiratory response survives a battery of falsification tests — placebo conditions with no pollution channel are unaffected, and fires that arrive only after an admission do not predict it.

This combination of a validated first stage, a dry-day respiratory response, and clean placebos leads us to interpret the FirePM25 estimates as fire-driven. The magnitude is modest on a typical fire day but rises steeply on the heaviest-smoke days in the upper decile of the exposure distribution, where most of the particulate burden is concentrated.

Our work builds on the vast empirical literature in economics that indicates a relationship between air pollution and health indicators. Burke et al. (2021) train a statistical model that relies on satellite fire and smoke data and ground pollution monitors to estimate that recent wildfires in the West Coast of the US have accounted for up to 25% of $\text{PM}_{2.5}$ in recent years in the country, and up to 50% in the westernmost regions of the US. Deryugina et al. (2019) estimate the causal effects of acute fine particulate matter exposure on mortality, health care use, and medical costs among the US elderly, exploiting the changes in local wind direction as an instrumental variable for pollution. Heft-Neal et al. (2019) show that this relationship holds even when the source of pollution is distant in relation to the observed outcomes, leveraging exogenous variation in local particulate matter exposure across sub-Saharan Africa due to dust export from the Bodele Depression, a remote Saharan region responsible for a substantial share of global atmospheric dust. Schlenker and Walker (2016) use the change in daily airport runway congestion, caused by network delays, to show that the local pollution levels in California affects contemporaneous health indicators of residents living nearby and downwind from airports. Closest to our setting, Rangel and Vogl (2019) use daily fire and wind variation to identify the effects of sugarcane-harvest smoke on health at birth in Brazil, and Rocha

and Sant’Anna (2020) document the local health consequences of Amazon fires. We complement this literature by tracing fire-driven $\text{PM}_{2.5}$ over more than 1,800 miles and recovering its health footprint at the far end of that transport.

This report contributes to the ongoing discussion about regulating deforestation by documenting additional health externalities of fires in the Amazon. As noted above, unchecked deforestation in the Amazon is expected to cause the biome to transition from a tropical rain forest to a Savannah-like biome (Lovejoy and Nobre, 2019). This transition has global implications as it would eliminate one of the world’s most important carbon sinks, further accelerating the arrival of excess deaths and economic costs associated with climate change. We provide added incentive to this, and the finding by Rocha and Sant’Anna (2020) on local effects, to reinstate the enforcement mechanisms that so effectively reduced deforestation and which may yet preserve the Amazon.

The rest of this paper is organized as follows: Section 2 discusses the causes and physical properties of Amazon fires; Section 3 provides a description of the data used in our approach; Section 4 explains the empirical strategy used to estimate the effects; Section 5 discusses the results of the paper; Section 6 assesses the robustness of estimated results; and Section 7 presents concluding remarks.

2 Background

Drivers of Amazon Fires and Physical Properties

Forest-clearing is one of the the main cause of fires in the Amazon Rainforest (Marle et al., 2017; Andreae, 1991). Forest-clearing fires begin at the end of the rainy season (May to beginning of June) when the moisture in the wood starts to evaporate, and peak in the dry season (August to November) when burning efficiency is higher (Andreae, 1991; Procopio et al., 2003).

Tropical deforestation fires release pollutants, primarily $\text{PM}_{2.5}$ such as black carbon (BC) and organic carbon (OC), in to the atmosphere, in the form of burned biomass aerosols (BBA) (Longo et al., 2013; van der Werf et al., 2010). The emission of $\text{PM}_{2.5}$ during the fires dramatically changes the chemical composition of the Amazon atmosphere

during the dry season. The concentration of aerosols and trace gases can increase by up to 8 times, affecting local natural atmospheric processes and the ecological equilibrium of the forest (Artaxo et al., 2005).

Baars et al. (2012) show that the aerosol composition in the Amazon atmosphere is the result of a complex mix of PM from internal and external sources. The aerosol composition suffers influence from external sources such as pollution from the Manaus industrial region, Saharan Dust, and BBA from other regions in South America and from Africa (Talbot et al., 1990; Artaxo et al., 1994; Kuhn et al., 2010). However, such out-of-basin sources are only relevant in the wet season, when fire counts are lower and rains do not allow local aerosol loads into the atmosphere. In the dry season, local BBA can disperse more effectively dominating the atmospheric composition (Baars et al., 2012; Guyon et al., 2003).

BBA are lifted to higher layers of the atmosphere where they can be easily be carried elsewhere by winds (Freitas et al., 2007; Garstang et al., 1988). The strong convective activity over the forest and pyroconvection – a strong convective force resultant from the heating of air parcels around fire – pushes the smoke towards the upper troposphere. BBA are vertically transported and once in the free troposphere, they are easily transported great distances (Baars et al., 2012). Therefore, the impacts on the troposphere chemistry are not only local, but also regional and possibly global (Artaxo et al., 1994).

3 Data

Our empirical analysis is based on a daily panel dataset built from multiple publicly available sources, from 2013 to 2019. Those include data provided by the National Aeronautics and Space Administration (NASA), the National Oceanic and Atmospheric Administration (NOAA), and the Brazilian Unified Health System (DATASUS). To allow for a more precise analysis in the next steps of this research, we also gather data starting in 2005 for all sources.

3.1 Fire Spots

We obtain the location of daily fire spots in the Legal Amazon region for the period of 2005-2019 using NASA’s Fire Information for Resource Management System (FIRMS)(Giglio et al., 2013). FIRMS contains data from both the MODIS sensor and the more recent VIIRS sensor. We focus on the MODIS product, which is less granular than VIIRS, but encompasses more years, allowing the analysis to be expanded until 2005.

3.2 Pollution

To understand the local relation between fire spots and pollution by BBA, we rely on NASA’s Modern-Era Retrospective Analysis for Research and Applications, Version 2 (MERRA-2), which provides a complete source on pollution and atmospheric conditions (Gelaro et al., 2017). We overlap the MERRA-2 rasters with the FIRMS points to evaluate the BC and OC concentrations at the fire spot. Since the MERRA-2 pixel is bigger (approximately 50km x 50km at Amazon Forest) than the fire point from FIRMS, we understand that the $PM_{2.5}$ concentration at the fire spot might not be the real value which would be measured in loco. Rather, it is the spatial average concentration of $PM_{2.5}$ in the pixel the wildfire is contained.

We read black carbon from the MERRA-2 surface mass concentration field and organic carbon from the corresponding carbonaceous aerosol field, expressing both as surface concentrations in kg/m^3 before transport, so that the two species enter the analysis on a common physical footing.²

We validate the $PM_{2.5}$ measurements generated by the atmospheric transport model with the data from ground monitors. Different than the US, Brazil does not have an extensive network of monitors with easy-to-access data. The states’ environmental agencies are the entities responsible for implementing pollution monitoring stations and disclosing the information. Therefore, we are restricted by the availability of information

²The two MERRA-2 carbonaceous diagnostics are reported on different bases — a surface mass concentration and a vertically integrated column mass density — and we place organic carbon on the same surface-concentration basis as black carbon using the planetary-boundary-layer mixing height described in Section 4. We carry both species, and the composite index defined below, side by side throughout.

disclosed by states and, to our knowledge, only the state of São Paulo discloses a full set of particulate matter and other pollutants information gathered by its environmental agency CETESB (*Companhia Ambiental do Estado de São Paulo*).

3.3 Fire-attributable Fine Particulate Matter

The health literature measures exposure in terms of fine particulate matter, $PM_{2.5}$, rather than in terms of its individual carbonaceous constituents. To express fire emissions in those units, we combine black and organic carbon into a single fire-attributable index. Primary biomass-burning $PM_{2.5}$ is dominated by black carbon and by organic matter, the latter being the organic carbon mass scaled up by the non-carbon atoms (hydrogen, oxygen, nitrogen) bound in organic aerosol (Turpin and Lim, 2001). Using an organic-matter-to-organic-carbon ratio of 1.6 and a primary-mass fraction consistent with biomass-burning emission factors for tropical vegetation (Andreae, 2019; Turpin and Lim, 2001), we form

$$\text{FirePM}_{25} = \text{BC} + 2.05 \cdot \text{OC}. \quad (1)$$

We use Equation (1) as a transparent bridge from emitted carbon to fine particulate mass; it preserves the compositional information — BC and OC remain separately reported — while delivering a single exposure measure in interpretable $PM_{2.5}$ units. The sensitivity of our estimates to the two chemistry parameters embedded in the coefficient 2.05 is examined in Section 6.

3.4 Health Indicators

We incorporate data on health outcomes from the DATASUS for the states and municipalities in the South and Southeast of Brazil. We subset for daily hospitalizations from medical conditions that are primarily affected by air pollution concentration, such as respiratory, and heart related diseases. In addition to respiratory and cardiovascular, we add a “placebo”, that is a medical condition that should not have a direct relationship with $PM_{2.5}$. Following (Schlenker and Walker, 2016), we chose appendicitis as being relevant to that end. For the next steps, we consider including other placebo conditions such as bone fractures, and stroke.

3.5 Ancillary Meteorological Conditions

The atmospheric transport model we use in the our empirical approach requires the input of meteorological conditions to compute the trajectory paths of air parcels. It relies on the Global Data Assimilation System (GDAS1) generated by the US National Weather Service’s National Centers for Environmental Prediction, which contains global information on wind vector components, temperature, and humidity on a $1^\circ \times 1^\circ$ grid (an approximately 110 km-pixel at the equator level).

4 Empirical Strategy

The objective of our analysis is to disentangle the health outcomes and the seasonal and economic correlates of field burning in the Amazon, as well as to overcome concerns related to measurement error in environmental indicators. Since the drivers of fire variation are expected to be related to potential determinants of health, even controls for seasonality and weather would not fully eliminate concerns about biased estimates when using naive regressions. Additionally, we would like to isolate other sources of pollution from our estimates, identifying the exact effect of pollutant emission from fires in the Amazon basin. To overcome this issue, we rely on an atmospheric-transport model to build a fire-attributable exposure measure that is plausibly orthogonal to other determinants of health.

Our strategy follows three steps. First, we use data on fire spots to pin down their exact location, and extract the level of Organic Carbon and Black Carbon from the pixel in those coordinates. Second, we feed those pollutants into an atmospheric transport model to generate municipality-by-day-level exposure of pollution driven by the fires. Third, we examine whether this exposure affects air quality and health outcomes on municipalities in the South and Southeast Brazil, more than 1000km away from the Amazon.

In order to determine how fire-driven pollution emissions are spatially dispersed across Brazil, we turn to the Hybrid Single Particle Lagrangian Integrated Trajectory Model (HYSPLIT), an atmospheric transport model developed by the U.S. National Oceanographic and Atmospheric Administration (NOAA), also used by Hernandez-Cortes

and Meng (2020), with meteorological conditions from NOAA’s Global Data Assimilation System (GDAS1). The key here is that emissions are not restricted to the Amazon area and pollution dispersal may vary depending on meteorological conditions and topography. Therefore, HYSPLIT allows us to fully capture the complexity of pollution transport and has the advantage of being more reliable than other transport models when modeling pollution transport in long-range settings (EPA, US, 2015).

Regarding our HYSPLIT implementation, we initialize all runs from the initial height of 30m, corresponding to the canopy height, and latitude and longitude given by the fire spots data. We run forward particle trajectories starting at the hour the satellite captured the fire spot, rounded down. Each trajectory runs for 2 weeks, a duration long enough to ensure most emitted particles reach the South and Southeast regions. Because HYSPLIT does not explicitly account for particle decay, we apply half-life parameters from the atmospheric chemistry literature set at 24 hours (EPA, US, 2018). We conservatively set the boundary layer height at 2 km above the surface — about double the maximum height for different types of surface (de Santana et al., 2020) — assuming that a particle no longer contributes to surface pollution concentrations once it exits the planetary boundary layer. The same 2 km mixing height places the organic-carbon column on the surface-concentration basis used for black carbon, so that the two species, and the FirePM25 index of Equation (1), share common units.

To obtain municipality-by-day pollution exposure, we sum across HYSPLIT trajectories for each municipality and day. Then, we examine whether HYSPLIT-predicted air pollution exposure correlates with ambient air pollution. Since the MERRA reanalysis dataset on pollution is partly based on an atmospheric transport model for measurement, we turn to pollution data collected by ground monitors in order to avoid an endogeneity issue. Unfortunately, ground monitors are not widely-spread in the South and Southeast regions of Brazil, and only the state of São Paulo presents a relevant network of those. Therefore, we restrict our sample to the HYSPLIT trajectories that reach this state, and estimate the following:

$$\text{Ground Monitor}_{i,d,m,y}^{p'} = \alpha(\text{HYSPLIT Pollution}_{d,m,y}^p) + X_{d,m,y} + \sigma_m + \rho_{i,y} + \epsilon_{i,d,m,y} \quad (2)$$

where $\text{Ground Monitor}_{i,d,m,y}^{p'}$ is the level of $p' \in \{\text{PM}_{2.5}, \text{PM}_{10}, \text{NO}, \text{SO}_2\}$ measured by the ground monitor i in day d , month m and year y ; $\text{HYSPLIT Pollution}_{d,m,y}^p$ is the

HYSPLIT-predicted level of $p \in \{\text{Organic Carbon, Black Carbon, FirePM25}\}$ for day d , month m and year y ; $X_{d,m,y}$ is a vector of weather controls; and σ_m and $\rho_{i,y}$ are month and station-year fixed-effects, respectively. The FirePM25 regressor is the headline of this first stage: because it is a fine-particulate-mass measure in the same family of units as the monitor reading, its coefficient α is a calibration slope, in principle bounded in $[0, 1]$ — the fraction of source-pixel particulate mass that survives transport and registers at the receptor.³

Because the ground monitors capture ambient pollution from many sources, we do not expect the predicted pollution to explain all of the variation in the regression. However, a positive correlation between the two measures, for $\text{PM}_{2.5}$ and PM_{10} , provides reassurance that HYSPLIT-predicted pollution is being detected by those monitors. We also regress other types of pollutants (NO and SO_2), not related to forest fires, on the HYSPLIT-predicted pollution to ensure that we are capturing increases on PM-related pollutants only. Results for those regressions are shown in Tables 1 and 2. Table 1 reports the relationship in both unstandardized form (Panel A, where black carbon, organic carbon, and FirePM25 are placed on a common scale — each multiplied by 10^6 — so their coefficients are legible and directly comparable) and standardized form (Panel B, where all regressors are expressed in standard-deviation units). Table 2 reports the corresponding NO and SO_2 specificity check with the regressors standardized to standard-deviation units, exactly as in Panel B of Table 1. The estimates show a positive and highly significant relationship between the HYSPLIT-predicted pollution — whether measured as black carbon, organic carbon, or the composite FirePM25 index — and PM-related pollutants, and no significant relationship with NO and SO_2 , the pollutants not related to forest fires.

Finally, we examine the effect of the HYSPLIT-predicted pollution on health outcomes — specifically, the hospitalization by cardiovascular and respiratory diseases

³In the table the three pollution regressors are placed on a common display scale (each multiplied by 10^6) purely for legibility; this rescales the reported α by a constant and leaves its sign, precision, and the comparison across columns unchanged. The trajectory regressor also sums a decay-weighted source concentration over every trajectory point crossing the receptor, so the underlying α absorbs the source-to-receptor integration footprint and is not itself restricted to $[0, 1]$. The interpretable object is the implied receptor concentration at a typical fire day, $\alpha \cdot \bar{X}$, or the semi-elasticity at the sample mean reported below. The qualitative reading — a positive, precisely estimated calibration of monitored $\text{PM}_{2.5}$ on fire-driven FirePM25 — is unaffected.

Table 1: OLS Results for the Effect of Fires on Pollution — Sao Paulo Only (2026; solution 8.2 dedup aggregation; OC unit-corrected; unstandardized and standardized regressors)

	PM2.5			PM10		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: Unstandardized (PM2.5, PM10 in ug/m³; BC, OC, FirePM25 x 1e6)</i>						
Organic Carbon	1.623*** (0.243)			2.462*** (0.288)		
Black Carbon		26.296*** (4.179)			38.134*** (4.812)	
FirePM25			0.768*** (0.115)			1.164*** (0.136)
Num.Obs.	36605	36605	36605	118286	118286	118286
R2	0.289	0.289	0.289	0.417	0.417	0.417
RMSE	7.96	7.96	7.96	15.39	15.39	15.39
FE: month	X	X	X	X	X	X
FE: station [^] year	X	X	X	X	X	X
FE: year	X	X	X	X	X	X
<i>Panel B: Standardized regressors (z-scores)</i>						
Organic Carbon	0.055*** (0.008)			0.030*** (0.004)		
Black Carbon		0.052*** (0.008)			0.028*** (0.004)	
FirePM25			0.055*** (0.008)			0.030*** (0.004)
Num.Obs.	36605	36605	36605	118286	118286	118286
R2	0.289	0.289	0.289	0.417	0.417	0.417
RMSE	0.84	0.84	0.84	0.76	0.76	0.76
FE: month	X	X	X	X	X	X
FE: station [^] year	X	X	X	X	X	X
FE: year	X	X	X	X	X	X

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Avg. of raw dep. var.: PM2.5 = 16.453, PM10 = 30.987 ug/m³.

Table 2: OLS Results — Sao Paulo Only (Other Pollutants, 2026 revision; standardized regressors; specificity check)

	NO			SO2		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Standardized regressors (z-scores); dependent variable in standard-deviation units</i>						
Organic Carbon	-0.001 (0.002)			-0.001 (0.004)		
Black Carbon		-0.002 (0.002)			-0.001 (0.004)	
FirePM25			-0.001 (0.002)			-0.001 (0.004)
Num.Obs.	89423	89423	89423	33604	33604	33604
R2	0.553	0.553	0.553	0.192	0.192	0.192
RMSE	0.67	0.67	0.67	0.90	0.90	0.90
FE: month	X	X	X	X	X	X
FE: station^year	X	X	X	X	X	X
FE: year	X	X	X	X	X	X
Avg. of raw Dep. Var. (ug/m^3)	17.893	17.893	17.893	5.212	5.212	5.212

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

One regressor per column; SD-per-SD slopes, as in Table 1 Panel B.

— at the municipality level. Because the dependent variable is a non-negative daily count, we estimate the relationship by Poisson pseudo-maximum-likelihood (PPML, `fepois` in the `fixest` package) rather than transforming the outcome: PPML is the natural model for a count, requires no log or inverse-hyperbolic-sine transform, and is robust to heteroskedasticity, with coefficients read as semi-elasticities (proportional changes in the expected count). Let $Y_{h\ell d}$ be the count of hospitalizations for condition $h \in \{\text{respiratory, cardiovascular}\}$ in municipality ℓ on day d . We estimate

$$\mathbb{E}[Y_{h\ell d} | \cdot] = \exp\left(\underbrace{\beta z_{\ell d}^{\text{fp}}}_{\text{wet-day slope}} + \beta_D (z_{\ell d}^{\text{fp}} \times \text{Dry}_{\ell d}) + \mathbf{w}'_{\ell d} \theta + \phi_{\ell d}\right), \quad (3)$$

where $z_{\ell d}^{\text{fp}}$ is the within-sample-standardized municipality-day FirePM25 index; $\text{Dry}_{\ell d} = \mathbf{1}[\text{wet}_{\ell d} = 0]$ indicates that the arriving plume met no measurable rain ($\text{wet} = \mathbf{1}[\text{rainfall} > 0.1 \text{ mm}]$); and $\mathbf{w}_{\ell d} = (\text{wet}, \text{heavy}, \text{mixdepth_km}, \text{temp}, \text{temp}^2)$ is a destination-weather control block — a two-part rain control, the boundary-layer mixing depth, and a temperature quadratic — each formed as a FirePM25-weighted mean of the HYSPLIT trajectory-endpoint meteorology. The wet-day exposure slope is β , the dry-day slope is $\beta + \beta_D$, and the reported interaction is β_D . The fixed-effect term $\phi_{\ell d} = \gamma_{\text{month}(d)} + \delta_{\ell \times \text{year}(d)}$ collects a calendar-month effect and the full municipality $\times$ year interaction (“FE-0”), so

that β is identified from within-municipality, within-year, across-month-and-day variation. Because the municipality $\times$ year effect absorbs any municipality-year-constant quantity — including the annual population — the count model and the per-capita (rate) model deliver numerically identical coefficients, and we report counts. Estimation is on the smoke-positive municipality-days (those with positive FirePM25) in the seven South and Southeast states, daily, over 2013–2019, with standard errors clustered by municipality.

We argue that, conditional on the municipality $\times$ year and month fixed effects and the destination-weather block, the residual variation in fire-driven PM_{2.5} arriving over a municipality on a given day is not related to other determinants of respiratory and cardiovascular admissions, so that identification of the causal effect follows from that.

5 Results

Table 3 presents our main specification, the municipality-day PPML estimates of Equation (3), for respiratory (column 1) and cardiovascular (column 2) hospitalization counts. Each column reports the wet-day FirePM25 slope, the dry-day interaction, and the implied dry-day slope, all per within-sample standard deviation of the FirePM25 index; the destination-weather block and the municipality $\times$ year and month fixed effects are included throughout, and standard errors are clustered by municipality. The sample is the smoke-positive municipality-days of the South and Southeast regions from 2013 to 2019.

The effect of fire-driven PM_{2.5} on respiratory admissions is concentrated on dry days. On days when the arriving plume meets measurable rain the slope is small and statistically indistinguishable from zero (-0.0015), but the dry-day interaction is positive and the implied dry-day slope is $+0.0050$, distinguishable from zero at the one-percent level: a one-standard-deviation increase in dry-day FirePM25 raises respiratory admissions by about half a percent. This split is what the physical mechanism predicts, since rain scavenges particulates out of the arriving plume before they can be inhaled. For cardiovascular admissions the same pattern is present in sign but is weaker and not statistically distinguishable from zero (an implied dry-day slope of $+0.0025$). We therefore read the evidence as an acute respiratory response to fire-driven fine particulate matter,

concentrated on the dry days when the plume is not rained out, and find no comparable evidence of a contemporaneous cardiovascular effect.

Table 3: PPML S3/FE-0 — Effect of fire-driven $PM_{2.5}$ on respiratory and cardiovascular hospitalizations (municipality, daily, 2013–2019)

	Respiratory (1)	Cardiovascular (2)
FirePM25 (z) — wet-day slope	-0.0015 (0.0033)	-0.0048 (0.0033)
× Dry day	+0.0064 (0.0042)	+0.0072 (0.0050)
FirePM25 (z) — dry-day slope	+0.0050** (0.0018)	+0.0025 (0.0023)
Num. Obs.	754,875	764,087
Avg. of Dep. Var.	0.805	0.900
Two-part rain controls	X	X
Mixing depth + temp. quadratic	X	X
Dry-day interaction	X	X
FE: month + municipality×year	X	X

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

PPML (`fepois`) on daily counts; smoke-positive municipality-days, 2013–2019; SE clustered by municipality.

Under municipality×year FE the count and per-capita (rate) PPML coincide (population absorbed).

Taken together with the first stage, the evidence tells a single, consistent story. The first-stage validation establishes that the HYSPLIT-derived FirePM25 index is detected by São Paulo ground monitors as genuine fine particulate matter, and not as some other pollutant; the municipality specification then shows that this fire-driven $PM_{2.5}$, once it has arrived over a municipality, raises respiratory hospital admissions on precisely the dry days when particulates are not scavenged from the air. As we show in Section 6, the same exposure leaves a battery of placebo conditions unaffected and does not predict admissions from fires that arrive only after the hospitalization, so we interpret the respiratory estimate

as the health footprint of long-range fire-driven $\text{PM}_{2.5}$.

6 Robustness

In order to check the robustness of our result to changes in model specification, we conduct two different sets of placebo tests to assess the possibility that the estimated effect is due to chance: a set of placebo *outcomes* with no pollution channel, and a placebo on the *timing* of exposure.

First, we re-estimate our main specification, Equation (3), for diagnoses unrelated to pollution. We replace the single appendicitis series of the original paper with a three-way battery, all built from the DATASUS-SIH microdata by ICD-10 chapter: **Placebo-K** is the whole K-chapter (digestive-system admissions — every ICD-10 code beginning with K, which contains appendicitis but is an order of magnitude denser); **Placebo-S** is the whole S-chapter (injury and trauma); and **Placebo-K+S** sums the two. These outcomes have no pollution channel and should not respond to fire-driven $\text{PM}_{2.5}$. Table 4 reports the PPML S3/FE-0 estimates for the three placebos. The implied dry-day slope — the object that carried the respiratory effect in Table 3 — is small and statistically indistinguishable from zero for every placebo chapter (+0.0016, +0.0018, and +0.0017), confirming that the dry-day response we estimate for respiratory conditions does not appear for conditions with no plausible link to air pollution.

As a second check, we test the timing of the exposure. The pollution must travel from the Amazon to the South and Southeast, and this transport takes time. Table 5 reports the distribution of the time-since-origin of HYSPLIT trajectory endpoints by state, computed from 14-day trajectories generated for a random ten-percent sample of 2019 fire spots: parcels reach the seven states with a mean lag of roughly eight to ten days and a standard deviation of about three days, defining a state-specific transport window of $[\text{mean} \pm 1 \text{ s.d.}]$ days.

The timing placebo turns this window around. Instead of assigning a municipality the pollution from fires that *precede* an admission by the transport window, we assign it the pollution from fires that occur in the symmetric *lead* window — fires that arrive only *after* the admission — which the transport logic says cannot have caused it. Concretely,

Table 4: PPML S3/FE-0 — Placebo outcomes (municipality, daily, 2013–2019). Placebo-K = whole ICD-10 K-chapter (digestive); Placebo-S = whole S-chapter (injury); Placebo-K+S = their sum.

	Placebo-K (1)	Placebo-S (2)	Placebo-K+S (3)
FirePM25 (z) — wet-day slope	-0.0093* (0.0045)	-0.0055+ (0.0028)	-0.0077* (0.0036)
× Dry day	+0.0110+ (0.0057)	+0.0073* (0.0031)	+0.0094* (0.0042)
FirePM25 (z) — dry-day slope	+0.0016 (0.0020)	+0.0018 (0.0017)	+0.0017 (0.0014)
Num. Obs.	758,365	731,402	777,267
Avg. of Dep. Var.	0.791	0.564	1.355
Two-part rain controls	X	X	X
Mixing depth + temp. quadratic	X	X	X
Dry-day interaction	X	X	X
FE: month + municipality×year	X	X	X

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

PPML (**fepois**) on daily counts; smoke-positive municipality-days, 2013–2019; SE clustered by municipality.

Under municipality×year FE the count and per-capita (rate) PPML coincide (population absorbed).

Table 5: Distribution for HYSPLIT trajectory points within each state

	SP	SC	RS	RJ	PR	MG	ES
N	37276	10098	22416	3446	30325	48686	3104
Mean	8.57	8.66	8.22	9.30	8.28	9.07	9.86
Std	3.09	3.18	3.29	2.75	3.22	2.94	2.40
Min	0.83	1.12	1.12	2.54	0.96	0.96	2.71
25%	6.08	6.08	5.54	7.125	5.67	7.0	8.0
50%	8.71	8.92	8.25	9.67	8.42	9.29	9.96
75%	11.21	11.29	11.04	11.46	10.96	11.42	11.75
Max	14	14	14	14	14	14	14

we re-estimate the main respiratory outcome by PPML on the full municipality-day panel, regressing the daily count on the log of the lead-window pollution, one column per indicator (the number of fires, organic carbon, black carbon, and the composite FirePM25 index), with the destination-weather block and dry-day interaction, month and municipality $\times$ year fixed effects, and standard errors clustered by municipality and year. Because the regressor is future pollution, every coefficient should be null. Table 6 confirms this: none of the lead-window exposure slopes — wet-day or dry-day, for any indicator — is distinguishable from zero at conventional levels, exactly as the timing logic requires. Hospitalizations respond to fires that precede them by the transport window, not to fires that follow.

Table 6: PPML days-before placebo — **municipality**-level, daily, 2013–2019. Future-fire pollution (lead transport window) on respiratory hospitalizations; should be null.

	(1) Number of Fires	(2) Organic Carbon	(3) Black Carbon	(4) FirePM25
Log(pollution), lead window — wet-day	+0.0060 (0.0056)	+0.0015 (0.0045)	+0.0010 (0.0044)	+0.0015 (0.0044)
× Dry day	-0.0038 (0.0030)	-0.0022 (0.0021)	-0.0021 (0.0023)	-0.0022 (0.0021)
Log(pollution), lead window — dry-day	+0.0022 (0.0048)	-0.0007 (0.0040)	-0.0011 (0.0042)	-0.0007 (0.0040)
Num. Obs.	7,216,784	7,216,784	7,216,784	7,216,784
Weather controls (wet, heavy, temp, temp ²)	X	X	X	X
FE: month + municipality $\times$ year	X	X	X	X

+ p<0.1, * p<0.05, ** p<0.01, *** p<0.001. PPML on daily counts; SE clustered by municipality & year.

Since neither placebo produces a relevant effect, we treat the dry-day respiratory effect as genuine.

Table 7: Sensitivity of B2 PM2.5 beta to chemistry assumptions

BC:OC ratio	OM:OC = 1.4	OM:OC = 1.6	OM:OC = 1.8
BC:OC = 1:5	1.372	1.208	1.079
BC:OC = 1:8	0.874	0.768	0.685
BC:OC = 1:12	0.589	0.517	0.461

Sensitivity to the chemistry of the FirePM25 index

The composite FirePM25 index of Equation (1) embeds two atmospheric-chemistry parameters: the black-to-organic carbon emission ratio and the organic-matter-to-organic-carbon factor that scales organic carbon into total organic aerosol mass. To show that our conclusions do not hinge on the particular values we adopt, we re-estimate the first-stage calibration of the FirePM25 index over a grid that varies the black-to-organic ratio across $\{1:5, 1:8, 1:12\}$ and the organic-matter factor across $\{1.4, 1.6, 1.8\}$, spanning the range reported for tropical biomass burning (Andreae, 2019; Turpin and Lim, 2001). The grid is reported in Table 7. The sign and statistical significance of the first-stage FirePM25 calibration slope are unchanged in every cell of the grid; only its magnitude moves, ranging over roughly a factor of three (from about 0.46 to 1.37) as the organic-carbon multiplier that builds the index is varied across its threefold range. The health specification is insulated from this rescaling by construction, because FirePM25 enters Equation (3) as a within-sample standardized (z -score) regressor, and its black-carbon and organic-carbon components carry nearly identical standardized signal in the first stage (Table 1, Panel B); reweighting them within the documented chemistry range therefore leaves the standardized slope, its sign, and its significance essentially unchanged. Our findings are thus robust to the chemistry parameters within the ranges the atmospheric-science literature supports.

7 Conclusion

In this paper, we explore whether the pollution generated by fires in the Brazilian Amazon affects health outcomes in the South and Southeast of Brazil, a region that is 1,800 miles south of the Amazon Rainforest and that contains the largest portion of the Brazilian population.

In order to reliably assess the long range health impacts of fires in the Amazon Rainforest, we used HYSPLIT, an atmospheric transport and dispersion model that offers hourly resolution air parcel trajectories. Our model enables us to disentangle the $\text{PM}_{2.5}$ exposure from Amazon fires from other sources of health determinants. In that way, we are able to determine the precise impact of pollutant emissions from Amazon forest fires on health, expressed through a single fire-attributable $\text{PM}_{2.5}$ index whose magnitude is directly interpretable.

We find that fire-driven $\text{PM}_{2.5}$ from biomass burning in the Amazon is detectable in the South and Southeast of Brazil and is associated with respiratory hospital admissions. The HYSPLIT-predicted FirePM25 index calibrates positively and precisely against ground-monitor $\text{PM}_{2.5}$ and PM_{10} in São Paulo; and, estimating Poisson regressions on daily municipality admission counts, fire-driven $\text{PM}_{2.5}$ raises respiratory admissions on the dry days when the arriving plume is not rained out, by about half a percent for a one-standard-deviation increase in exposure, with no comparable effect on cardiovascular admissions. The effect is modest on a typical fire day but rises steeply on the heaviest-smoke days in the upper decile of the exposure distribution, where the particulate burden is concentrated. We provide evidence that such $\text{PM}_{2.5}$ effects are limited to conditions related to air-pollution exposure: a battery of placebo conditions is unaffected, fires that follow an admission do not predict it, and pollutants unrelated to biomass burning are uncorrelated with the index.

In line with the previous literature on the health outcomes of $\text{PM}_{2.5}$, our study generates enough evidence to show that research that ignores long-range transport may underestimate the costs of deforestation. We hope to be adding to the ongoing discussion of the health externalities of tropical-forest deforestation. Our results provide added incentive to reinstate the enforcement mechanisms that effectively curbed deforestation, preserving the Amazon and human lives.

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A Appendix

A.1 Contemporaneous tables under FE-C

As a counterpart to the within-municipality-year (FE-0) fixed effects of our main specification, Tables 8 and 9 report the same S3 PPML specification of Equation (3) under the additive **FE-C** structure, $\phi_{\ell d} = \gamma_{\text{month}(d)} + \lambda_{\text{year}(d)} + \delta_{\ell}$ (month + year + municipality). FE-C loses almost no observations to Poisson separation and is therefore the near-full-sample counterpart of FE-0. The headline is materially unchanged: the respiratory dry-day slope is +0.0039 and remains distinguishable from zero, the cardiovascular slope remains indistinguishable from zero, and the placebo dry-day slopes stay small and insignificant.

Table 8: Appendix A.1 — PPML S3/FE-C (municipality, daily). FE-C = month + year + municipality (additive).

	Respiratory (1)	Cardiovascular (2)
FirePM25 (z) — wet-day slope	-0.0014 (0.0033)	-0.0043 (0.0035)
× Dry day	+0.0053 (0.0044)	+0.0065 (0.0054)
FirePM25 (z) — dry-day slope	+0.0039* (0.0018)	+0.0022 (0.0025)
Num. Obs.	791,981	792,265
Avg. of Dep. Var.	0.805	0.900
Two-part rain controls	X	X
Mixing depth + temp. quadratic	X	X
Dry-day interaction	X	X
FE: month + year + municipality	X	X

+ p<0.1, * p<0.05, ** p<0.01, *** p<0.001

PPML (`fepois`) on daily counts; smoke-positive municipality-days, 2013–2019; SE clustered by municipality.

Under municipality×year FE the count and per-capita (rate) PPML coincide (population absorbed).

Table 9: Appendix A.2 — PPML S3/FE-C placebo outcomes (municipality, daily).

	Placebo-K	Placebo-S	Placebo-K+S
	(1)	(2)	(3)
FirePM25 (z) — wet-day slope	-0.0089+	-0.0054+	-0.0074+
	(0.0049)	(0.0029)	(0.0038)
× Dry day	+0.0104+	+0.0059+	+0.0085+
	(0.0060)	(0.0033)	(0.0045)
FirePM25 (z) — dry-day slope	+0.0014	+0.0005	+0.0011
	(0.0019)	(0.0018)	(0.0014)
Num. Obs.	792,092	791,570	792,332
Avg. of Dep. Var.	0.791	0.564	1.355
Two-part rain controls	X	X	X
Mixing depth + temp. quadratic	X	X	X
Dry-day interaction	X	X	X
FE: month + year + municipality	X	X	X

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

PPML (`fepois`) on daily counts; smoke-positive municipality-days, 2013–2019; SE clustered by municipality.

Under municipality×year FE the count and per-capita (rate) PPML coincide (population absorbed).

Table 10: B1: Standardized BC and OC — Sao Paulo Only (2026; OC unit corrected)

	PM2.5 (z-scored)		PM10 (z-scored)	
	(1)	(2)	(3)	(4)
Organic Carbon (z-score)	0.055*** (0.008)		0.030*** (0.004)	
Black Carbon (z-score)		0.052*** (0.008)		0.028*** (0.004)
Num.Obs.	36605	36605	118286	118286
R2	0.289	0.289	0.417	0.417
RMSE	0.84	0.84	0.76	0.76
FE: month	X	X	X	X
FE: station $\hat{\text{year}}$	X	X	X	X
FE: year	X	X	X	X
Avg. of Dep. Var. (raw)	16.453	16.453	30.987	30.987

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

A.2 Standardized regressors in the first stage

The main-text first-stage tables report black and organic carbon in raw concentration units, where the two species differ in scale by orders of magnitude and their coefficients are not directly comparable. As an alternative presentation, Tables 10 and 11 re-estimate the first-stage validation with each regressor standardized to zero mean and unit variance, so that every coefficient is the response to a one-standard-deviation increase in the regressor and the black-carbon and organic-carbon slopes can be read side by side. The standardized first-stage slopes on black carbon, organic carbon, and the composite index are nearly identical (around 0.055 standard deviations of monitored PM_{2.5} per standard deviation of predicted pollution), confirming that the compositional split carries essentially the same signal as the single FirePM25 index.

A.3 Aerosol optical depth as an alternative exposure measure

The FirePM25 index is built from surface and column carbonaceous-mass fields, which require an assumption about the mixing height to place them on a common basis.

Table 11: B1: Standardized BC and OC — Other Pollutants (2026)

	NO (z-scored)		SO2 (z-scored)	
	(1)	(2)	(3)	(4)
Organic Carbon (z-score)	-0.001 (0.002)		-0.001 (0.004)	
Black Carbon (z-score)		-0.002 (0.002)		-0.001 (0.004)
Num.Obs.	89423	89423	33604	33604
R2	0.553	0.553	0.192	0.192
RMSE	0.67	0.67	0.90	0.90
FE: month	X	X	X	X
FE: station $\hat{\text{year}}$	X	X	X	X
FE: year	X	X	X	X
Avg. of Dep. Var. (raw)	17.893	17.893	5.212	5.212
+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001				

A cleaner alternative would build the exposure measure from aerosol optical depth, an intrinsically column-integrated quantity with a standardized mapping to surface $\text{PM}_{2.5}$ (van Donkelaar et al., 2010). Constructing this measure requires a large satellite-reanalysis extraction that is outside the scope of the present draft; we flag it as the most promising robustness extension and leave it for future work.